

Change the Subject: Subject–Object Contrasts in Counterfactuals

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Abstract

If Socrates resembled Adonis, they would both be handsome. If Adonis resembled Socrates, neither would be handsome. And if Adonis and Socrates resembled each other, they might both be handsome and they might not. This is a puzzle for theories of conditionals that predict ‘if A , would C ’ and ‘if B , would C ’ to be equivalent whenever A and B have the same truth conditions. This paper evaluates four responses to these data:

I show that this puzzle cannot be solved by appealing to *de re* readings of the comparator, and offer a solution by developing a semantics of conditionals that is sensitive to what conditional antecedents are about.

Goodman (1947) observed that logically equivalent sentences can behave differently as counterfactual antecedents. He noticed that ‘New York City is in Georgia’—the Southern US state, as opposed to the Caucasian nation—is true in all the same worlds as ‘Georgia includes New York City’, but they come apart as counterfactual antecedents; similarly for ‘I am Julius Caesar’ and ‘Julius Caesar is me’.¹

- (1) a. If New York City were in Georgia, then New York City would be in the South.
- b. If Georgia included New York City, then New York City would be in the South.

(1a) seems true, while (1b) does not.

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¹This differs slightly from Goodman’s (1947, 121) original example, in which both sentences are intuitively true and have incompatible consequents. However, they leave open the reply that one sentence is *vacuously* true. I have reformulated the example to avoid the issue. Goodman also offers the example of counteridenticals (*if I were Julius Caesar* versus *if Julius Caesar were me*), which have spawned their own rich literature (Lakoff 1996; Kocurek 2018; Kaufmann 2018) and are certainly relevant, but which I do not discuss here due to space constraints.

Examples like Goodman’s are important because they appear to challenge Substitution, the widely-endorsed principle of counterfactual logic that whenever two sentences are logically equivalent—true in all the same worlds—they are intersubstitutable *salva veritate* in counterfactual antecedents.²

Substitution. For any sentences p and q , if p and q are logically equivalent (true in all the same worlds), then for any sentence r , ‘if p , would r ’ and ‘if q , would r ’ are logically equivalent too.

This paper has two goals. First, I show that contrasts like Goodman’s examples are no isolated edge case. They are far more widespread than we might initially have thought. In particular, I show that similar examples arise for expressions that have previously been argued to exhibit a figure–ground asymmetry. They appear, for example, with binary relations (*near*, *resembles*, *is similar to*), comparatives (*taller than*), kinship terms (*sister*, *cousin*, *teammate*), prepositions (*on top of*, *beneath*), and quantifiers (*every*, *no*). They are so commonplace that any theory of counterfactuals ought to say something about them.

Second, I critically evaluate four possible responses to these data.

1. *The de re reply.* The third is to deny that the counterfactual antecedents in question are logically equivalent. One way to do this is to appeal to the *de re/de dicto* ambiguity. According to this approach, by default, subjects in counterfactual antecedent are interpreted *de dicto*, while objects are interpreted *de re*. Thus, for example, (1a)’s antecedent expresses the proposition is true at a world w just in case the location of New York City *in w* is within the location of Georgia *in the actual world*, while (1b)’s antecedent expresses the proposition that is true at a world w just in case the location of Georgia *in w* includes the location of New York City *in the actual world*. Since these propositions are not logically equivalent, on this reading, (1) is not a counterexample to Substitution after all.
2. *The strict counterpart reply.* The second is Lewis’s reply to Goodman’s New York–Georgia pair. Lewis (1973b, 43) suggests that subject terms summon up a less stringent counterpart relation than object terms.

I present a variety of challenges for this view. In particular, I show that it struggles to account for antecedents of the form ‘if Alice were taller than she is’, which would require the counterpart relation to be loose and stringent at the same time.

3. *The fine-grain reply.* The first is to replace worlds with something more fine-grained. In particular, Fine (2012) develops a truthmaker semantics of conditionals, based not on possible worlds but on (partial) states, and an exact verification relation between states

²Substitution is also known as Left Logical Equivalence, or LLE (Arló-Costa 2007). Chellas (1975) calls it RCEA, ‘the Rule of Classically Equivalent Antecedents.’ It is validated by the theories of Stalnaker (1968), Lewis (1973b), Chellas (1975), Kratzer (1981), Pollock (1976), Burgess (1981), Bennett (1984), and Delgrande (1987). Exceptions include Nute (1980), Fine (2012), Ciardelli, Zhang, and Champollion (2018), and Santorio (2018).

and sentences. Fine’s theory allows conditionals with logically equivalent antecedents to differ in their truth value, provided their antecedents do not have the same verifiers.

While this has the resources to account for the data we consider, it comes at a cost. I show that doing so requires making controversial assumptions about which states exist (assumptions that Fine himself rejects).

This approach account for our data using independently-attested readings (the *de re/de dicto* ambiguity). However, I argue that it faces serious challenges. Among them: there are appear to be sentences that behave differently as counterfactual antecedents even on their *de dicto*, logically equivalent readings; and it violates plausible, independently-attested observations about the *de re/de dicto* ambiguity.

4. *The aboutness reply.* The fourth is claim that the antecedents to make counterfactual sensitive to a notion of *aboutness*, over and above truth conditions, whereby two sentences may be true in all the same worlds but differ in what they are about. On this approach, our selection of counterfactual possibilities is guided not only by the antecedent’s truth conditions, but also what the sentence is *about*. The fundamental idea is that we interpret a counterfactual antecedent, we change the part of the world its antecedent is about. This idea may be interpreted in different ways: that aboutness features in the semantics of counterfactuals, or instead in a merely pragmatic constraint guiding a context-dependent accessibility relation.

I show that it has the resources to account for our data, and moreover overcomes the challenges facing the truthmaker, strict counterpart, and *de re* replies. At the same time, as the newest of the four approaches, it leaves room for future development; in particular, regarding how to determine the aboutness relation.

It is important to be clear on what the data we consider do *not* show. I do not believe that they uniquely determine a correct theory of counterfactuals. In a field as rich as counterfactuals, this would be hard for any data to do. However, they do reveal some approaches to be more plausible than others. While my sympathies lie with the aboutness approach, the ultimate question of which theory of counterfactuals to adopt must be settled by considerations beyond this paper.

The paper proceeds as follows. In Section 1, ...

1 Subject and Object Contrasts in Counterfactuals

In *The Principles of Mathematics*, Russel distinguishes what he calls the *subject* and *assertion* of sentences. Comparatives illustrate the distinction: “In a relational proposition, say ‘*A* is greater than *B*,’ we may regard *A* as the subject and ‘is greater than *B*’ as the assertion, or *B* as the subject and ‘*A* is greater than’ as the assertion. There are thus, in the case proposed, two ways of analyzing the proposition into subject and assertion” (Russell 1903, §48).

Russell did not apply his observations about comparative statements—such as *A is greater than B*—to counterfactuals. When we do so we find some stark contrasts.

- (2) *Alice is 25, Bob is 15. One must be over 21 to enter the bar.*
- a. If Alice were younger than Bob, they could both enter the bar.
 - b. If Bob were older than Alice, they could both enter the bar.

We naturally read the first as false and the second as true. However, Alice is younger than Bob just in case Bob is older than Alice. When we interpret “If Alice were younger than Bob” we have a default tendency to imagine changing Alice’s age and leaving Bob’s fixed: Alice’s age is the figure, Bob’s age the ground. Similarly, when we interpret “If Bob were older than Alice” we tend to imagine changing Bob’s age while leaving Alice’s fixed. It appears that each sentence privileges a particular division of figure and ground, just as we do when we adopt a particular interpretation of Rubin’s vase. For this reason, I will use the term *figure–ground contrasts* for contrasts like those in (2), though I intended this to be a neutral label, without it communicating any substantive commitments about the exact nature of the contrast.

Figure–ground contrasts seem to arise for comparatives in general. For example:

- (3) *Alice is 1.5m tall, Bob is 1m tall. One must be at least 1.2m tall to ride the Ferris wheel.*
- a. If Alice were shorter than Bob, they could ride the Ferris wheel together.
 - b. If Bob were taller than Alice, they could ride the Ferris wheel together.

1.1 Prepositions

Talmy (1975, 2000) observed that a similar contrast appears not only for comparatives, such as *A is younger than B*, but for a large class of transitive verbs, such as *A is near B*. For Talmy, this is contrast between *figure* and *ground*. Like Russell, who argued that *humanity belongs to Socrates* and *Socrates is human* express distinct propositions, Talmy proposed that *the bike is near the house* and *the house is near the bike* “in fact do not say the same thing” (Talmy 1975, 420). In the former, the house “has a set location within a framework [...] and is to be used as a reference-point by which to characterize the other object’s (the bike’s location), understood as a variable,” whereas the latter makes all the reverse specifications. Talmy does not discuss reciprocals such as *the bike and house are near each other*, though these also appear to exhibit a different division of figure and ground compared to *the bike is near the house* and *the house is near the bike*, with the figure consisting of both objects.

Counterfactuals are sensitive to this division of figure and ground. For example:

- (4) *Alice is at home and dry. Bob is outside in the rain.*
- a. If Alice were near Bob, they would both be dry.
 - b. If Bob were near Alice, they would both be dry.
 - c. If Alice and Bob were near each other, they would both be dry.

We naturally judge the first to be false, the second to be true, and the third to be indeterminate. Though Alice is near Bob just in case Bob is near Alice, which holds just in case they are near

each other.³ The same pattern holds for a wide variety of relations, such as *next to*, *beside*, and *adjacent to*.

- (5) *Alice lives in an expensive area and works in an affordable area.*
- If Alice's house were adjacent to her office, her rent would be higher.
 - If Alice's office were adjacent to her house, her rent would be higher.
 - If Alice's office and house were adjacent, her rent would be higher.

All sentences whatsoever that express spatial comparisons appear to exhibit figure–ground contrasts: *near*, *above/below*, *to the left/right of*, and so on. Suppose that three blocks, A, B, and C, are placed in a grid as in Figure 1, where each block occupies exactly one square. Consider (6).

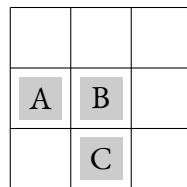
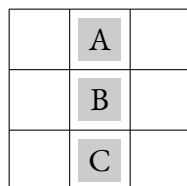


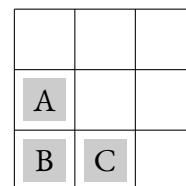
Figure 1

- (6) a. If block A were on top of block B, the blocks would form a straight line.
b. If block B were beneath block A, the blocks would form a straight line.

The first is most naturally judged true and the second false. Figure 2 depicts the arrangements we tend to imagine when we interpret each antecedent.



(a) If block A were on top of block B, ...



(b) If block B were beneath block A, ...

Figure 2

1.2 Non-spatial relations

Talmy observes that non-spatial relations also express a figure–ground distinction. Take resemblance: “the sentence ‘she resembles him,’ which might be thought to derive from something like ‘she is near him in appearance, or her appearance is near his appearance,’ is not understood in the same sense as ‘he resembles her’ ” (Talmy 1975, 421). In conditionals this contrast surfaces

³See Dowty (1991, 556) and Gleitman et al. (1996) for arguments that *near* is truth-conditionally symmetric.

as a difference in truth conditions. For example, Socrates was not known for being handsome, possessing a snub nose and bulging eyes (Plato, *Theaetetus* 143e). Adonis, on the other hand, was famously handsome. With that in mind, consider (7).

- (7) a. If Socrates resembled Adonis, they would both be handsome.
- b. If Adonis resembled Socrates, they would both be handsome.
- c. If Adonis and Socrates resembled each other, they would both be handsome.

The first is intuitively true, the second false, and the third indeterminate.

We see an analogous contrast with *similar*.

- (8) a. If Bach were similar to Black Sabbath, Bach would be played less at church.
- b. If Black Sabbath were similar to Bach, Bach would be played less at church.
- c. If Bach and Black Sabbath were similar, Bach would be played less at church.

1.3 Relational nouns

While Russell and Talmy do not consider relational nouns (such as *compatriot*, *sister*, and *colleague*), they give rise to similar contrasts in counterfactual antecedents.

- (9) a. If Bizet had been Verdi's compatriot, they would both have been Italian.
- b. If Verdi had been were Bizet's compatriot, they would both have been Italian.
- c. If Bizet and Verdi had been compatriots, they would both have been Italian. (after Quine 1950)
- (10) *Amy has harsh parents, Beth has nice parents. Each person is happy just in case she has nice parents.*
 - a. If Amy were Beth's sister, they would both be happy.
 - b. If Beth were Amy's sister, they would both be happy.
 - c. If Amy and Beth were sisters, they would both be happy.
- (11) *Alice works for a bank. Bob works as a teacher. Alice hates her job and wants to be a teacher. Bob likes his job.*
 - a. If Alice were Bob's colleague, they would both like their jobs.
 - b. If Bob were Alice's colleague, they would both like their jobs.
 - c. If Alice and Bob were colleagues, they would both like their jobs.

In each example, the first counterfactual is intuitively true, the second false, and the third indeterminate.

1.4 Quantifiers

Hempel (1945) famously observed that *Every raven is black* is true just in case *Everything that is not a raven is not black* is, but they seem to differ in what observations support them. Spector and

Mouly (2025) notice that quantifiers also give rise to sentences that appear truth-conditionally equivalent but behave differently as counterfactual antecedents. Imagine a simple universe each object must be either a raven or a parrot, and either black or white. Thus in this universe there are four possible kinds of object: a black raven, a white raven, a black parrot, and a white parrot (Figure 3).



Figure 3

Consider (12), for example, in the context of Figure 3.

- (12) a. If every raven were black, there would be three black birds.
 b. If every white bird were a parrot, there would be three black birds.
 c. If every bird were either a parrot or black, there would be three black birds.

Intuitively, the first is true, the second is false, and the third is indeterminate. When we suppose that every raven is black, we tend to change the ravens and leave the non-ravens as they are. While when we suppose that every white bird a parrot, we tend to change the white birds and leave the non-white birds as they are. And when we suppose that every bird is either a parrot or black, we may imagine the white raven changing into a parrot or to black (perhaps alongside further changes).

The point applies broadly. *No A is B* and *No B is A* have the same truth conditions, as do *Some A is B* and *Some B is A*. But they come apart in counterfactual antecedents.

- (13) a. If no raven were white, there would be three black birds.
 b. If no white bird were a raven, there would be three black birds.
 (14) a. If some white raven were black, there would be three black birds.
 b. If some black bird were a white raven, there would be three black birds.

In the context of Figure 3, intuitively the first sentence in each pair is true and the second false.

One might object that our knowledge of the world—say, our knowledge that black ravens are more plentiful than white ravens—impacts our judgements in these cases. In reply, consider a more clinical setting, based on one by Spector and Mouly. Imagine a simple universe in which each object must be either a square or circle, and either light or dark. Consider four shapes, arranged as in Figure 4.

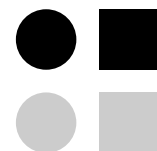
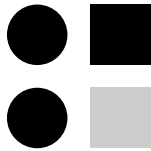


Figure 4

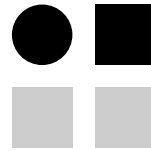
“Every circle is dark” has the same truth conditions as “every non-dark object is not a circle,” which in this limited universe is equivalent to “every light shape is square.”

But these sentences intuitively give rise to different possibilities as conditional antecedents (see Figure 5).

- (15) a. If every circle were dark, there would be three dark shapes.
 b. If every light shape were square, there would be three dark shapes.
 c. If every shape were either square or dark, there would be three dark shapes.



(a) If every circle were dark, ...



(b) If every light shape were square, ...

Figure 5

1.5 Beyond Counterfactuals

Figure–ground contrasts are not limited to counterfactuals. We observe analogous contrasts for desire verbs like *want* and *wish*. This is unsurprising, given the popularity of analyses of desire verbs that appeal to counterfactuals (e.g. Stalnaker 1984; Heim 1992; Grano and Phillips-Brown 2022; Wimmer 2024).

- (16) *Decorators are moving furniture according to their client's instructions. Currently, the bed is on the left and the painting is on the right (see Figure 6). The client wants both to be on the left. The manager instructs the employee:*
- a. The client wants the painting to be above the bed.
 b. The client wants the bed to be below the painting.
- (17) *The client is looking at their bedroom (Figure 6). They say:*
- a. I wish the painting were above the bed.
 b. I wish the bed were below the painting.
- (18) *It's so cold in New York City!*
- a. I wish New York City was in Georgia.
 b. I wish Georgia included New York City.

In (16), (17) and (18), the first sentence of each pair is intuitively true, while the second is not.

The same applies to causal claims. This is again unsurprising, given the popularity of counterfactual approaches to causation (Lewis 1973a; Collins, Hall, and Paul 2004).

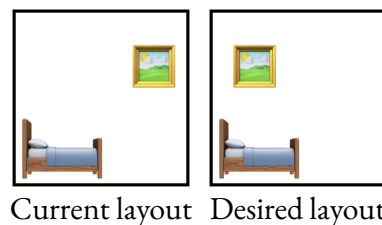


Figure 6

(19) *Georgia has a warmer climate than New York City, allowing Georgia to grow tasty peaches.*

- a. New York City grows bad peaches because it's not in Georgia.
- b. New York City grows bad peaches because Georgia does not include it.

(19a) is intuitively true while (19b) is not (for the latter to be true, we would require, say, that the state of Georgia has agricultural policies that promote growing good peaches, so that if New York City were where it currently is but under Georgia's jurisdiction, it would grow good peaches). Similarly, (20a) is intuitively true while (20b) is not.

(20) *The client inspected the current layout and complained that the movers did not follow the instructions.*

- a. The fact that the painting is not above the bed caused the client to complain.
- b. The fact that the bed is not below the painting caused the client to complain.

These examples show that figure–ground contrasts are not specific to the conditional construction itself—say, to the presence of *if*. They rather arise in the presence of counterfactual meaning, broadly construed.

1.6 The empirical phenomenon

Before examining how to account for these data, it is worth stating a descriptive generalisation to capture what we have observed. Here is a first shot:

(21) When we suppose a counterfactual antecedent featuring a subject and an object, we tend to allow the subject's properties to vary and hold fixed the object's properties.⁴

By 'fixing x 's properties', I mean that x has the same intrinsic properties in every hypothetical world we consider as x has in the actual world. We need the restriction to intrinsic properties to rule out extrinsic properties such as 'not resembling Socrates'. If such extrinsic properties were allowed, changing Socrates' appearance would inadvertently count as changing Adonis's properties, which is not what we want.

Here is a simple formalisation. Let f be a counterfactual selection function taking a sentence, a world, and perhaps other parameters (such as an accessibility relation, a similarity relation) and returning a set of worlds. Intuitively, $f(A, w)$ is the set of worlds that result from supposing the sentence A true at world w . Given a class of properties C (for example, the class of intrinsic

⁴Strictly speaking, it would be more appropriate to speak of subject and object *terms* that *refer* to entities; that 'Socrates' is the subject term and 'Adonis' is the object term, and that the subject term refers to Socrates and the object term refers to Adonis. I use the more concise formulation above for simplicity, with the understanding that this more exact formulation can be recovered if needed.

properties), we *fix x 's properties* (with respect to C, f, A , and w) just in case for every property P in C and world w' in $f(A, w)$, x has property P in w' if and only if it has property P in w . We allow x 's properties to vary (with respect to C, f, A , and w) just in case we do not fix x 's properties (with respect to C, f, A , and w).

The generalisation in (21) is formulated in terms of counterfactual antecedents. This neglects the wide range of expressions we have considered that exhibit similar contrasts, such as desire verbs and causal claims. For this reason, let us speak not of supposing a counterfactual antecedent, but instead more neutrally of supposing a *sentence*. I assume that when we interpret a desire verb or a causal claim, we perform this act of supposition; for example, that interpreting ‘The client wants the painting to be above the bed’ requires supposing ‘the painting is above the bed’.

- (22) When we suppose a sentence featuring a subject and an object, we tend to allow the subject's properties to vary and hold fixed the object's properties.

While this generalisation is on the right track, it has an obvious flaw. In many cases, supposing the supposing the antecedent true is not compatible with holding the object's properties fixed. One example comes from the passive voice.

- (23) a. If Zara hugged Cyril, he would blush.
b. If Cyril was hugged by Zara, he would blush.

For Cyril to be hugged by Zara, Zara's properties must change (she must move in the appropriate way to hug Bob), even though she appears as the object in ‘Cyril is hugged by Zara’.

- (24) *In a chess match, Polgár won against Anand.*
a. If Anand had won against Polgár, Anand would have gotten the trophy.
b. If Polgár had lost against Anand, Anand would have gotten the trophy.

For Anand to win against Polgár, Polgár's properties—specifically, her property of having actually won the match—must change, even though she appears as the object in ‘Anand won against Polgár’.

- (25) *Alice and Bob are sitting on opposite ends of a see-saw. Currently Alice is low and Bob is high.*
a. If Alice were higher than Bob, Alice's legs would be dangling in the air.
b. If Bob were lower than Alice, Alice's legs would be dangling in the air.

Given the contextual fact that Alice and Bob are sitting on opposite ends of a see-saw (and that the see-saw is rigid, and doesn't break, and so on), for Alice to be higher than Bob, Bob's position must change, even though he is the object in ‘Alice is higher than Bob’.

- (26) a. If the age of retirement were 65, Alice could retire.
b. If 65 were the age of retirement, Alice could retire.

Given that 65 is a number (presumably having its properties essentially), it is not possible to change what 65 is, while it is easy to imagine the age of retirement having a different value.

The sentences in each pair from (23) to (26) are intuitively equivalent. The different divisions of subject and object do not affect the conditionals' truth value. In contrast, in the previous examples we considered, the subject–object division does affect truth conditions (such as *resembles*, *taller than*, and so on). In those cases, supposing the sentence was compatible with fixing the object's properties, allowing the default to apply. For instance, when we suppose that Socrates resembled Adonis, there is a selected world where Socrates has the appearance that Adonis actually has.

In light of these examples, our rule must include a proviso that holding the object's properties fixed be compatible with supposing the sentence—whether in virtue of the antecedent's truth conditions alone (such as what it means to hug someone) or also contextual factors (such as the fact that Alice and Bob are on a see saw). In terms of our simple formalisation using conditional selection functions, we may express this as the requirement that $f(A, w)$ contain at least one world where we hold the object's properties fixed.

- (27) When we suppose a sentence featuring a subject and an object, we tend to allow the subject's properties to vary and hold fixed the object's properties, provided that this is compatible with supposing the sentence.

This rule has a striking feature: it does not care about *which* properties we fix. The only constraint is that they be the *subject's* properties, or the *object's* properties. For example, when we interpret 'if Socrates resembled Adonis', the rule in (27) tells us to allow Socrates' properties to vary. It does not tell us more specifically whether we to vary his appearance, age, height, or other properties.

In many cases this is not a problem. General features of counterfactual interpretation will settle which properties to vary. For example, when we interpret 'if Alice were older than Bob', we vary Alice's age and do not vary her height. On a similarity approach, for example, this simply be because the most similar worlds to the actual world where Alice is older than Bob are worlds where she has the same height that she actually has.

It is, however, a problem for quantified statements (examples (12)–(15) above), such as 'if every circle were dark'. Intuitively, when we interpret this, we take the shapes that are actually circles, and change their colour but not their shape. In theory, an alternative way for 'every circle is dark' to come out true is to turn any light circles into squares. But this is not what we observe.⁵ As Spector and Mouly (2025) observe, predicates appearing in the sentence's restrictor (circle) are fixed, while the predicates appearance in the sentence's scope (is dark) may vary. This gives us the following rule.

- (28) When we suppose a sentence featuring a subject and an object, we allow the subject's properties appearing in the sentence's scope to vary and hold fixed the object's properties, provided that this is compatible with supposing the sentence.

⁵I am grateful to Benjamin Spector for discussion of this point.

Let us now consider how to account for this phenomenon.

2 The fine-grain reply

The first reply we consider is that figure–ground contrasts in counterfactuals call for a more fine-grained approach to meaning. According to possible worlds semantics (Lewis 1970; Partee 1977), the meaning of a sentence—such as a counterfactual antecedent—can be modelled as the set of possible worlds in which it is true. This framework automatically validates Substitution (which, recall, is the principle that whenever p and q are true in all the same worlds, then the counterfactuals ‘ $\text{if } p \text{ would } r$ ’ and ‘ $\text{if } q \text{ would } r$ ’ are too, for any consequent r).

One might propose that the fault lies with the coarse grain of truth conditions, and that by moving to a more fine-grained, hyperintensional notion of meaning, in which logically equivalent sentences can be distinguished, the overall architecture of one’s theory of conditionals might be maintained.

For instance, Stalnaker (1968) introduces a selection function that takes a proposition (a set of possible worlds) and a possible world and returns a single world. One might think that if the selection function took as input a *sentence*, rather than a *proposition*, his theory can account for the figure–ground contrasts in counterfactuals we have observed.

The problem, however, runs deeper than that. Substitution follows from quite weak principles in the logic of conditionals, such as Weaker than Entailment and Reciprocity.⁶

Weaker than Entailment. If A entails C , then *if* A , *would* C is true.⁷

Reciprocity. If $A > B$ and $B > A$ are true, then $A > C$ is true if and only if $B > C$ is true.

For if A and B are logically equivalent, they entail one another, so by the Weaker than Entailment, $A > B$ and $B > A$ are both true. Then by Reciprocity, $A > C$ is true just in case $B > C$ is. Thus no matter how fine-grained we take meanings to be, provided our theory of conditionals validates Weaker than Entailment and Reciprocity, it will automatically validate Substitution too.

⁶The name Reciprocity comes from Egré and Rott (2021). Nute (1980) calls it CSO, and Starr (2022) calls it Substitution of Subjunctive Equivalents. A wide variety of theories of conditionals validate both of these principles, including those of Stalnaker (1968) and Lewis (1973b). Stalnaker regards Reciprocity as essential to his theory, remarking that “the selection is based on an ordering of possible worlds with respect to their resemblance to the base world. If this is correct, then [Reciprocity] must be imposed [...]. These conditions on the selection function are necessary in order that this account be recognizable as an explication of the conditional” (Stalnaker 1968, 36). For a proof that Lewis’s account validates Reciprocity, see McHugh and Kłochowicz 2024, 171.

⁷Weaker than Entailment in turn follows from plausible principles such as Identity ($A > A$) and Right Weakening (if B entails C then $A > B$ entails $A > C$). For a discussion of semantics that—surprisingly—invalidate Identity, see Mandelkern (2021).

2.1 Fine

Fine (2012) proposes a semantics of counterfactuals that does not validate Substitution. It has three primitive notions: (i) a set of *states*, representing parts of the world, (ii) an *exact verification* relation between states and sentences, telling us which states exactly verify which sentences, and (iii) a *transition relation* between states, stating when one state is a ‘possible outcome’ of another at a given world.

He offers the following semantics, a counterfactual *if A would C* is true at a world *w* just in case for every exact verifier *t* of *A* and possible outcome *u* of *t* at *w*, *u* contains an exact verifier of *C* (Fine 2012, 237).

Following Fine (2014b, 576), let us call two sentences *exactly equivalent* just in case they have the same exact verifiers and falsifiers. Two sentences may be logically equivalent without being exactly equivalent; for example, *A*, $A \vee (A \wedge B)$, and $(A \wedge B) \vee (A \wedge \neg B)$ are logically but not exactly equivalent. Fine’s semantics therefore does not validate Substitution, but it does satisfy an analogous principle.

Exact Substitution. If *A* and *B* are exactly equivalent, then $A > C$ is true if and only if $B > C$ is true.

This theory has the resources to account for the figure–ground contrasts in counterfactuals, provided that the antecedents in question are not exactly equivalent. Given that the state space and exact verification relation are primitives of Fine’s theory, whenever we have two antecedents *p* and *q* which behave differently as counterfactual antecedents, it is consistent with Fine’s theory to stipulate that *p* and *q* do not have the same exact verifiers. Recall (6), for example, showing that ‘Block A is on top of block B’ and ‘Block B is beneath block A’ behave differently in counterfactual antecedents. To account for this on Fine’s theory, we may stipulate that the state space contains a state exactly verifying that block A is on top of block B, without exactly verifying that block B is beneath block A.

Such a stipulation is certainly possible, but it is plausible? According to Fine (2000), it is not.

Suppose that a given block *a* is on top of another block *b*. Then there is a certain state of affairs *s*₁, we may describe as the state of *a*’s being on top of *b*. There is also a certain state of affairs *s*₂ that may be described as the state of *b*’s being beneath *a*. Yet surely the states *s*₁, and *s*₂ are the same. There is a single state of affairs *s* “out there” in reality, consisting of the blocks *a* and *b* having the relative positions that they do; and the different descriptions associated with *s*₁, and *s*₂ would merely appear to provide two different ways at getting at this single state of affairs.

(Fine 2000, 3)

This strongly suggests that *Block a is on top of block b* and *Block b is beneath block a* have them same exact verifiers.⁸ Given this, (6) is a counterexample to Exact Substitution.

⁸Geach and Williamson propose a related, though not identical view. Geach states that “a relation neither exists

For a second example, recall from (5) that ‘A is adjacent to B’ and ‘B is adjacent to A’ behave differently as counterfactual antecedents. According to Fine, however, “the state of a ’s being adjacent to b is surely the same as the state of b ’s being adjacent to a ” (2000, 17), which would imply that the sentences in (5) are equivalent on Fine’s theory of counterfactuals.

While Fine appeals to intuition to support his claims, there are positive arguments that speak in their favour. Here are two.

First, it is hard to understand what such a state could be, that verifies that block A is on top of B, without exactly verifying that block B is beneath A. One suggestion is that it is the the state of block B being in a particular location L, such as the bottom left corner of Figure 1. Such a suggestion, however, conflicts with plausible principles governing exact verification. It is widely assumed that states necessitate the truth of the statements they exactly verify: if s exactly verifies A then necessarily, whenever s obtains, A is true (Fine 2017c). However, the state of block B being in L does not necessitate that that B is beneath A. For in a world where B is in L but A is not above L, the state will obtain but B will not be beneath A. It seems that we do not have any independent grasp on states could have the required properties.

Second, the proposal seems to require necessitation relations between states whose nature is also mysterious. One may wish to maintain that ‘A is on top of B’ are logically equivalent ‘Block B is beneath block A’, despite not being exactly equivalent (their logical equivalence, after all, would motivate moving to a more fine-grained framework in the first place). Within work on truthmaker semantics, it is commonly thought that contingent statements—such as ‘Block A is on top of B’ and ‘B is beneath A’—are true at a world just in case the world contains an exact verifier of the statement (in other words, that whenever a statement is true at a world, the world loosely verifies the statement; see Fine 2014b, 4, Fine 2017a). Given this, if ‘A is on top of B’ and ‘B is on top of A’ are logically but not exactly equivalent, every world containing an exact verifier of one must also contain an exact verifier of the other, despite the verifiers being distinct: the presence of an exact verifier of one must necessitate the presence of an exact verifier of the other. One may stipulate such a necessitation relation between set of states, but the need for additional stipulations weakens the account’s explanatory power.

For such a stipulation to be plausible, one would ideally want to have an independent grasp of this necessitation relation. To my knowledge, truthmaker semantics provides only one such necessitating relation: *parthood*. Whenever a state obtains, its parts do too. However, using parthood, the account would require every that exact verifier of ‘A is on top of B’ have as a proper part an exact verifier of ‘B is beneath A’, and vice versa. This, in turn, would imply the existence of an infinite descending chain of states, $s_1 > t_1 > s_2 > t_2 > \dots$, where each s_i exactly verifies ‘A is on top of B’ and each t_i exactly verifies ‘B is beneath A’.⁹ While this position is consistent,

nor can be observed apart from its converse relation” (Geach 1957, 33), and Williamson argues that relations are identical to their converses: “‘--- stabs ...’ and ‘--- is stabbed by ...’ (and ‘... stabs ---’) stand for the same relation” (Williamson 1985, 249). For discussion see MacBride (2007), Gaskin and Hill (2012), and Liebesman (2013).

⁹This is because, given that $\ulcorner$ if p , would $r \urcorner$ is true and $\ulcorner$ if q , would $r \urcorner$ is false, on Fine’s theory there must be a state t_1 that exactly verifies q but not p . And given that $\ulcorner$ if q , would $\neg r \urcorner$ is true and $\ulcorner$ if p , would $\neg r \urcorner$ is false, there

it seems extremely implausible that accounting for simple examples like (6) should require us postulate the existence of such an infinite descending chain of states.

To sum up, while Fine’s theory of counterfactuals can account for the figure–ground contrasts above, it does so at the cost of some highly controversial assumptions about the exact verification relation, which Fine himself would likely reject. It would require the existence of states, and necessitation relations between them, of which we seem to have no independent grasp. For this reason, it is worth exploring alternative analyses of figure–ground contrasts in counterfactuals.

3 The *De Re* Reply

The first reply we consider is that object terms are read *de re* by default. I call this the *de re* reply.

Aristotle observes that “It is possible to walk while sitting” has both a consistent and a contradictory reading (*Sophistical Refutations* 166a24–166a30; see Dutilh Novaes 2003). Nowadays we would label the consistent reading as *de re*, and analyse it as saying that there is a possible world w where someone who is sitting in the actual world w_0 is walking in w . The contradictory reading is the *de dicto* one: there is a possible world w where someone who is sitting in w is walking in w .

De re/de dicto ambiguities also appear in comparatives. In ‘On Denoting’ Russell recalls the following anecdote.

I have heard of a touchy owner of a yacht to whom a guest, on first seeing it, remarked, “I thought your yacht was larger than it is”; and the owner replied, “No, my yacht is not larger than it is.” What the guest meant was, “The size that I thought your yacht was is greater than the size your yacht is”; the meaning attributed to him is, “I thought the size of your yacht was greater than the size of your yacht.”

(Russell 1905, 489)

The same ambiguity appears in conditional antecedents (Lewis 1973b, 37; Kennedy 1995; Heim 2001; Borisov 2016; Wehmeier 2012).

(29) If your yacht were larger than it is, you would boast about it more.

This has a consistent reading, in which we imagine the yacht being larger in the hypothetical worlds than it is in the actual world. Compare it with (30).

(30) If your yacht were larger than itself, you would boast about it more.

The antecedent is inconsistent. A plausible idea is that (29) admits a *de re* reading, while (30) only admits a *de dicto* reading (plausibly, forced by *itself* being bound by *your yacht*).

must be a state s_1 that exactly verifies p but not q . Then if s_1 and t_1 are distinct, and t is part of s_1 , t_1 must be a proper part of s_1 . By similar reasoning, there is an exact verifier s_2 of p that is a proper part of t_1 , an exact verifier t_2 of q that is a proper part of s_2 , and so on.

Quantifier restrictors also have *de re* readings (Percus 2000).

- (31) a. Alice thinks that everyone inside the room is outside the room.
b. If everyone inside the room were outside the room, the room would be empty.

The belief claim and antecedent have a consistent reading, saying that everyone who is inside the room *in the actual world* is outside the room in the worlds, respectively, compatible with Alice's beliefs and raised by the conditional antecedent.

Lewis (1973b, 37) represents the different readings via quantifier scope. He translates *If your yacht were larger than it is, ...* as *The size of your yacht is an x such that, if the size of your yacht exceeded x , ...*. This results in the right reading for Russell's example but not for (31b). This sentence does not say that everyone in the room is an x such that, if x were outside the room, it would be empty. On this reading we remove one person at a time and the sentence is false, whereas when we interpret (31b) we remove them all at once and the sentence is true.

An alternative is to represent *de re–de dicto* readings using world indices (Percus 2000; Keshet 2010).¹⁰ Assuming a simple, Hintikka (1962) style semantics of belief for expository purposes, we can represent the two readings as follows, where w_0 is the world of evaluation.

- (32) I thought your yacht was larger than it is.
a. **De dicto.** For every world w compatible with my beliefs at w_0 , the size of the yacht at w is greater than the size of the yacht at w .
b. **De re.** For every world w compatible with my beliefs at w_0 , the size of the yacht at w is greater than the size of the yacht at w_0 .¹¹
- (33) Alice thinks that everyone inside the room is outside the room.
a. **De dicto.** For every world w compatible with Alice's beliefs at w_0 and for all x such that x is inside the room at w , x is outside the room at w .
b. **De re.** For every world w compatible with Alice's beliefs at w_0 and for all x such that x is inside the room at w_0 , x is outside the room at w .

One may argue that the figure–ground contrasts we have considered here exhibit a similar ambiguity. For example, “Socrates resembles Adonis” may express the proposition that is true at a world w just in case Socrates' appearance in w is sufficiently close to Adonis's appearance *in the actual world* w_0 , and that “every raven is black” may express the proposition that is true at a world w just in case every raven in w_0 is black in w .

This *de re* reading appears to be available. Suppose Alice believes that Bob resembles the actor Remi Malek. Alice isn't aware that Remi has an identical twin brother Sami Malek—she

¹⁰I use world indices since they allow us to conveniently express all the relevant readings, though nothing I say in what follows hinges on this choice.

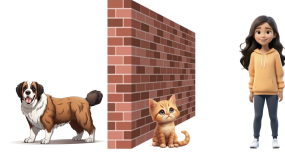
¹¹This reading differs slightly from Russell's own gloss of his example: “The size that that I thought your yacht was...”. As Kripke (2005) notes, Russell assumes that there was a particular size the guest believes the yacht to be. (32b) does not require this.

has no idea that Sami exists.

(34) Alice believes that Bob resembles Sami Malek.

I submit that, in the right context, (34) may have a true reading.¹²

Or suppose Suzy can see the cat but not the dog (Figure 7). She has no idea that the dog is there. Actually, the cat is near the dog. Consider:



(35) Suzy believes that the cat is near the dog.

This is most naturally read as false—after all, Suzy has no beliefs whatsoever about the dog. But I submit that it may also have a true reading in this context, saying that Suzy believes that the cat is near the location where the dog actually happens to be. On this *de re* reading, (35) is true.

The *de re* reply asserts that *de re* readings are responsible for the contrasts we have observed. Recall, for example, (2).

- (2) *Alice is 25, Bob is 15. One must be over 21 to enter the bar.*
- a. If Alice were younger than Bob, they could both enter the bar.
 - b. If Bob were older than Alice, they could both enter the bar.

On the *de re* reading these are equivalent, respectively, to (36).

- (36) a. If Alice were younger than 15, they could both enter the bar.
b. If Bob were older than 25, they could both enter the bar.

The antecedents in (2) are equivalent *de dicto* but not *de re*. Read *de re*, they are not counterexamples to Substitution and can easily be handled by extant semantics of counterfactuals, such as Stalnaker's or Lewis's.

To account for the contrasts we observed using *de re* readings, we may say that counterfactual antecedents are interpreted *de re* by default.

While I believe that the required *de re* do exist, it faces significant challenges.

¹²Granted, this reading is extremely fragile. It seems to disappear as soon as Alice has any beliefs whatsoever about Sami Malek. For instance, if Alice knows that Sami Malek exists, but has no idea what he looks like, (34) no longer has any true reading. One explanation for this is that (34), on its *de dicto* reading, presupposes that Alice believes that Sami Malek exists, but does not presuppose this on its *de re* reading. The *de re* reading emerges as a last resort to rescue the sentence from having a false presupposition. Thus once its presuppositions are met, only the *de dicto* reading is available.

3.1 Challenges for the *de re* reply

3.2 Challenge 1: *de dicto* antecedents that violate Substitution.

The first and most serious challenge to the *de re* reply is that there are sentences which behave differently as counterfactual antecedents even on their *de dicto* readings.

For example, suppose there are two points, A and B, sitting on a horizontal plane (see Figure 8). Point A is currently located at 2 and point B is currently located at 4. Thus their midpoint is currently located at 3. Since the location of their midpoint depends on the locations of A and B, if A or B moves, their midpoint moves accordingly. Now consider (37).

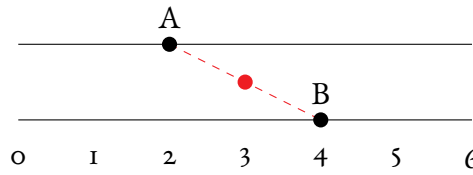


Figure 8: Point A is at 2, point B is at 4, so their midpoint is at 3.

(37) *Point A is at 2, point B is at 4, so their midpoint is at 3 (Figure 8).*

- a. If A were to the right of A and B's midpoint, A would be to the right of 4.
- b. If B were to the left of A and B's midpoint, A would be to the right of 4.

Intuitively, (37a) has a true reading here. On this reading, we imagine A's location changed and B where it is. Since A is in a different location, their midpoint must be too. And given that B is located at 4, the only way for A to be to the right of their midpoint is for A to be to the right of 4.¹³

This is the *de dicto* reading: it asks us to imagine A moving to the right of where their their midpoint *would be*, not where it actually is. On this reading, "A is to the right of their midpoint" expresses the proposition that is true at a world *w* must in case A's location in *w* is to the right of the location of their midpoint in *w*.

While (37a) has a true reading, (37b) does not appear to have any true reading whatsoever. However, given that (37a) has a true *de dicto* reading, we would expect (37b) to have one too. And crucially, on their *de dicto* readings, the antecedents of (37a) and (37b) are logically equivalent.¹⁴ (37) therefore contains sentences that are truth conditionally equivalent on their *de dicto* readings, but behave differently as counterfactual antecedents—contrary to the *de re* reply.

¹³(37a) also has a false/indeterminate reading on which it asks us to imagine A to the right of where their midpoint *actually* is: 3. In that case, A could be at 3.5, say, in which case it would not be to the right of 4. While this reading certainly exists, the crucial observation is that (37a) has a true reading.

¹⁴It suffices to note that A is to the right of A and B's midpoint just in case A is to the right of B; that is, $a > \frac{a+b}{2}$ if and only if $a > b$.

3.3 Challenge 2: object terms in counterfactual antecedents are not interpreted *de re* by default.

According to the *de re* reply, objects in counterfactual antecedents are interpreted *de re* by default. A wide range of cases show this to be incorrect.

For a first case, take the antecedent “if A were higher than B”. According to the *de re* reply, when we interpret this we tend to imagine A higher than B *currently* is. To test this claim, suppose that the left side of a see-saw is at its lowest point, touching the ground, while the right side is at its highest point, one metre off the ground (see Figure 9).

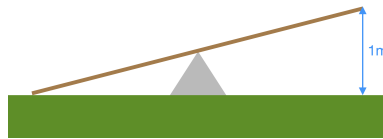


Figure 9

Compare the antecedents:

- (38) a. If the left side were higher than the right side
b. If the left side were higher than the right side is

These differ in an interesting way. (38a) is naturally interpreted *de dicto*, asking us to imagine a world where the left side in that world is higher than the right side in *that very same* world. In contrast, (38b) is naturally interpreted *de re*, asking us to imagine a world where the left side in that world is higher than the right side in the *actual* world. The difference gives rise to different judgements for counterfactuals.

- (39) a. If the left side were higher than the right side, ...
 (i) the left side would be more than one metre off the ground. ✗
 (ii) the left side would be at most one metre off the ground. ✓
 b. If the left side were higher than the right side is, ...
 (i) the left side would be more than one metre off the ground. ✓
 (ii) the left side would be at most one metre off the ground. ✗

The expression “if A were higher than B is” shows us how a bone fide *de re* reading behaves. It is not how “if A were higher than B” behaves. This suggests that “if A were higher than B” is most naturally not read *de re*.¹⁵

For a second example, suppose A and B are two rubber ducks floating in a tank of water. The water level in the tank can be raised or lowered, which determines how high A and B are. Consider:

¹⁵A remaining possibility is that “if A were higher than B is” is obligatorily *de re*, while “if A were higher than B” has both a *de re* and a *de dicto* reading. However, this would predict (37a) and (37b) to have a *de dicto* reading on which they are equivalent, which we do not observe. It further faces Challenges 3–5 below.

(40) If A were higher than B currently is, A and B would still be at the same height.

(40) is easily judged true. This judgement comes about from reasoning that, if A were higher than B currently is, the water level would be higher, in which case A and B would still be at the same height (to strengthen this reading, we imagine, say, a movable rod running from the tank walls to the ducks, forcing them to take sit at the water level). Compare this with (41).

(41) If A were higher than B, A and B would still be at the same height.

(41), in contrast, is contradictory.

Further evidence comes from conjunctive antecedents such as (42).

(42) If Adonis resembled Socrates and Socrates were handsome, they would both be handsome.

(42) is most naturally judged true. It is true on the *de dicto* reading of “Adonis resembles Socrates” but false on the *de re* reading. Read *de dicto*, the antecedent is true at a world *w* just in case Adonis’ appearance at *w* is sufficiently like Socrates’ appearance at *w* and Socrates is handsome at *w*. In this case, both are handsome and we therefore judge the conditional true. Read *de re*, the antecedent is true at a world *w* just in case Adonis’ appearance at *w* is sufficiently like Socrates’ appearance at the *actual world* *w*₀—in which Socrates is not handsome—and Socrates is handsome at *w*. In that case the consequent is false. The fact that (42) is easily judged true shows that “Adonis resembles Socrates” in (42) is most naturally read *de dicto*.

Lastly, consider examples like (43).

(43) *Lucy’s family are taking a family photo. Lucy is currently blocking the person behind her from being in the photo.*

- a. If Lucy were sitting next to where Lucy is currently sitting, everyone would be in the photo.
- b. If Lucy were sitting next to Lucy, everyone would be in the photo.

(43a) is easily judged true, whereas (43b) sounds contradictory. Or take (44).

(44) If every student were taller than every student, the school basketball team would be better.

(44)’s antecedent sounds contradictory. But its *de re* reading is consistent, asking us to imagine every student taller than the actual height of every student (i.e. taller than the current height of the tallest student).

These judgements challenge the *de re* reply. If objects are interpreted by default in counterfactual antecedents, as the *de re* reply alleges, it is mysterious why this default vanishes in many quite disparate cases. We might call this the ‘problem of the missing *de re*’. The problem is even more pressing in light of (41), (44), and (43b), since would expect the *de re* reading to be preferred

when the *de dicto* reading is contradictory. The *de re* reply fails to explain why the alleged *de re* readings of these sentences are unavailable.

3.4 Challenge 3: some terms seem to lack the required *de re* readings altogether

Suppose two blocks are sitting on a table. If block A were pushing against block B, where would the blocks be? The most natural answer, I believe, is that A would have moved to where B is, with B's location unchanged; vice versa for "if block B were pushing against block A". For the *de re* reply to account for this, we must assume that *pushing against* has a *de re* reading, on which "A is pushing against B" is true at a world w just in case A's location in w is pushing against B's location in the actual world w_0 .

This is a highly implausible proposal for what *pushing against* could mean. There appears to be no evidence that it has such a *de re* reading. Take belief reports—a reliable environment in which *de re* readings appear. Suppose that blocks A and B are at opposite ends of a table, as in Figure 10. Alice believes that block A is to the right of the table (in fact, where A would be, were it pushing against B), but has no beliefs about block B. Consider (45).

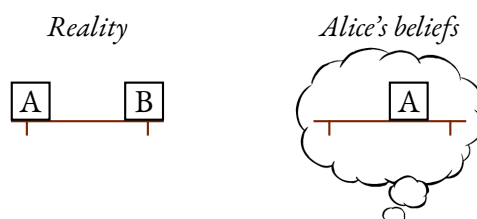


Figure 10

(45) Alice believes that block A is pushing against block B.

(45) does not appear to have any true reading here. However, according to the *de re* reply, 'A is pushing against B' should also admit a *de re* reading on which (45) says that in each of Alice's belief worlds w' , block A's location in w' is pushing against B's location in the actual world. This is true in the scenario above, while (45) has no true reading.

For a second case, recall from Section 1.3 that relational nouns like *sister* exhibit a subject–object contrast (examples). When we interpret *if A were B's sister*, we tend to imagine A and B having B's current parents rather than A's current parents. For the *de re* reply to account for this, we must assume that *sister* has a *de re* reading, on which "A is B's sister" is true at a world w just in case B's parents in the actual world are A and B's parents in w .

As before, there is no evidence that kinship terms have such *de re* readings. To illustrate with belief reports again:

- (46) *José believes that Alice's parents are Vivienne and Michael, and does not have any beliefs about who Beth's parents are. In fact, Vivienne and Michael are Beth's parents.*
 José believes that Alice is Beth's sister.

(46) does not appear to have any true reading in this scenario. Compare this with (47).

- (47) José believes that Beth's parents are Alice's parents.

(47) readily admits a *de re* reading on which it is true in the above scenario. This shows that there is nothing especially difficult about entertaining this *de re* meaning—it is just not something that (46) can express.

Summing up this section, for the *de re* reply to account for subject–object contrasts exhibited by *touching* and relational nouns like *sister*, it must posit implausible *de re* readings for which there appears to be no independent evidence.

3.5 Challenge 4: violations of the principle of charity

The fourth challenge for the *de re* reply is that it violates independently attested principles governing the availability of *de re* readings; in particular, it violates the principle of charity.

It is generally assumed that *de dicto* and *de re* readings are always in principle available (Percus 2000; Keshet and Schwarz 2019). We use pragmatic reasoning, contextual features and our general world knowledge to disambiguate. This has the attractive feature of making the resolution of *de re–de dicto* ambiguity of a kind with the resolution of any other ambiguity, such as referential ambiguity (for instance, determining who ‘they’ refers to in a given context) and polysemy (disambiguating riverbanks from financial banks). Grammar supplies both as interpretative possibilities, context disambiguates.

A pervasive feature of ambiguity resolution is that we prefer resolutions on which the sentence is true to those on which it is false. In Russell's yacht example, “I thought your yacht was larger than it is,” we opt for the *de re* reading. It is easy to understand why: the *de dicto* reading is clearly false. We can account for this behaviour by a principle of interpretative charity.¹⁶

Principle of Charity. When multiple readings of a sentence are available, some of which are true, we opt for one of the true readings.

To illustrate, suppose you see someone you know searching through a crowd. When you ask what they are doing here they reply, “I'm looking for a friend.” Here we easily interpret them as saying that there is a particular friend they are looking for. This is the *de re* reading.

¹⁶The Principle of Charity I have in mind here is one appearing in the work of Quine and Davidson. Wilson, who coined the term, wrote that “we act on what might be called the Principle of Charity. We select as designatum that individual which will make the largest possible number of [...] statements true” (Wilson 1959, 532). As Davidson puts it, the task of solving for meaning is aided by “assigning truth conditions to alien sentences that make native speakers right when plausibly possible” (Davidson 1973, 324; for discussion see Glüer 2011).

Now suppose you meet someone at a bar who tells you they are new to the neighbourhood and don't know anyone yet. When you ask what they are doing here they reply, "I'm looking for a friend." Now we naturally interpret them as saying that they are looking to have a friend. There is no particular person they are looking for. For all we know they might not even have any friends. This is the *de dicto* reading.

In each case, we seamlessly manage to pick the reading that makes the sentence true. If a *de re/de dicto* ambiguity were at play in the subject–object asymmetries we have considered, we would likewise expect each reading to be available, and that when the sentence is true on one reading and false on the other, we opt for the true reading.

This is not what we observe. To illustrate, recall (2), for example, showing that 'older than' and 'younger than' exhibit a subject–object contrast. According to the *de re* reply, 'A is younger than B' and 'B is older than A' have a *de dicto* reading on which they are equivalent. Given the principle of charity, we would expect this reading to be available when it rescues the sentence from falsity. Is it? Consider:

(48) *Alice is 25, Bob is 15. One must be over 21 to enter the bar.*

- a. If Alice were younger than Bob, it's possible that they could both enter the bar.
- b. If Bob were older than Alice, it's possible that they could both enter the bar.

For theories of conditionals that validate Substitution, (48a) and (48b) are equivalent on their *de dicto* reading. For instance, within the similarity approach (Stalnaker 1968; Lewis 1973b), it is plausible to assume that worlds where we change Alice's age and fix Bob's are as close, as all being equal, as worlds where change Bob's age and fix Alice's. Then (48a) and (48b), on their *de dicto* readings, should both be true.

However, there does not appear to be any reading on which (48a) is true. Rather, it seems straightforwardly false, and certainly not equivalent to (48b). Thus, if the *de re* reply were right, we would have to rethink plausible ideas about the availability of *de dicto* and *de re* readings; in particular, we would have to give up—or at least significantly curtail—the principle of charity.

3.6 Challenge 5: the required *de re* propositions might not exist

The fifth and final challenge for the *de re* reply is that the required *de re* propositions might not exist. The challenge comes from the meaning of *mistakenly believes*.

Here is a simple analysis of mistaken belief: a person *S* mistakenly believes a proposition *p* just in case (i) *S* believes *p* and (ii) *p* is false. On this view, the believed proposition and the false proposition are one and the same. Things might turn out to be more complicated than this (see Blumberg 2018 for some discussion). As a null hypothesis, however, this analysis seems quite reasonable.

Now consider the following case, illustrated in Figure 11 (I have labelled the locations 1, 2 and 3 for convenience).

- (49) *John believes that Alice is in the rain and Bob is dry. In fact both are in the rain.*
 John mistakenly believes that if Alice were near Bob, Alice would be dry.



Figure 11

(49) is intuitively true. Assuming a simple, Hintikka-style analysis of belief, (49) states that for each of John's belief worlds w at the actual world w_0 , Alice is dry in the selected worlds at w where the proposition p expressed by the antecedent is true. The question is what p is. There are relevant three possibilities.

1. *De re*, anchored to the actual world: if Alice were near where Bob actually is.
 $p = \{w' : \text{Alice's location in } w' \text{ is near Bob's location in } w_0\}$
2. *De re*, anchored to John's beliefs; if Alice were near where John believes Bob to be.
 $p = \{w' : \text{Alice's location in } w' \text{ is near Bob's location in } w\}$
3. *De dicto*; if Alice were near Bob.
 $p = \{w' : \text{Alice's location in } w' \text{ is near Bob's location in } w'\}$

Both *de re* readings predict (49) to be false, given our analysis of mistaken belief. On the first *de re* reading, (49) is false because, if Alice were near where Bob actually is (location 2), Alice would be wet—(i) fails. On the second *de re* reading (49) is false because, if Alice were near where John believes him to be (location 3), Alice would indeed be dry, so John wouldn't be mistaken—(ii) fails.

That leaves the *de dicto* reading. On this reading, “Alice is near Bob” is equivalent to “Bob is near Alice” (on its *de dicto* reading) and to “Alice and Bob are near each other”. For theories that validate Substitution, then, (49) is equivalent to (50a), on its *de dicto* reading, and to (50b).

- (50) a. John mistakenly believes that if Bob were near Alice, Alice would be dry.
 b. John mistakenly believes that if Alice and Bob were near each other, Alice would be dry.

Unlike (49), however, these are intuitively false, and not equivalent to (49).

Assuming our simple analysis of mistaken belief, the *de re* reply is not able to make sense of these judgements. It predicts there to be *no* reading of (49) on which it is true both and not equivalent to (50a) and (50b).

4 The counterpart reply

The second reply we consider comes from Lewis. He offered the following account of Goodman's New York–Georgia pair in (1).

We can explain the simultaneous truth of Goodman's sentences [...] by the hypothesis that both are *de re* both with respect to 'New York City' and with respect to 'Georgia', and that a less stringent counterpart relation is summoned up by the subject terms 'New York City' in [(1a)] and 'Georgia' in [(1b)] than by the object terms 'Georgia' in [(1a)] and 'New York City' in [(1b)]. Then in [(1a)] we are concerned with the closest worlds to ours where a not-too-close counterpart of our New York is in a close counterpart of our Georgia, and hence is in (a counterpart of?) the South; whereas in [(1b)] we are concerned with the closest worlds to ours where a not-too-close counterpart of our Georgia includes a close counterpart of our New York City, and hence is not entirely included in the South.

(Lewis 1973b, 43)

The general idea is that we interpret subject terms using a looser counterpart relation than object terms. Call this the *counterpart reply*.

The counterpart reply readily extends to the other contrasts we have observed. For example, in 'Adonis resembles Socrates', the subject term is 'Adonis' and the object term is 'Socrates'. According to Lewis, then, we would expect that Adonis's counterparts may have a different appearance from the actual Adonis, while Socrates's counterparts must have the same appearance as the actual Socrates.

4.1 Extending the counterpart reply into a viable theory

For the counterpart reply to serve as a viable theory, I believe it must be extended in two ways.

Firstly, we must take the counterpart relation to be loose or strict *with respect to a particular property*. To see why, consider (51).

(51) *Socrates is speaking with Diotima, while Adonis is in another town. Adonis is handsome while Socrates is not.*

- a. If Adonis were near Socrates and Socrates resembled Adonis, Diotima would gaze at them.
- b. If Socrates were near Adonis and Adonis resembled Socrates, Diotima would gaze at them.

Intuitively, (51a) is true and (51b) is false. To predict this on Lewis's approach, (51a) must summon up a loose counterpart relation with respect to Adonis's location but a strict counterpart relation with respect to his appearance, while for Socrates it is the reverse. In (51b) this is swapped.

Second, it must include the *de re* readings of comparators from the *de re* reply. This is motivated by examples like (52).¹⁷

(52) If Alice were taller than she is, she would have been picked for the basketball team.

Here Alice appears as both a subject and object. A naive application of Lewis's proposal would predict her to summon up both a loose and strict counterpart relation at once, which is impossible. And even if one overcomes this challenge, if there are two counterpart relations at play, there remains the issue of how to determine which counterpart relation should be used to interpret the occurrence of 'she' in the consequent.

We naturally interpret (52) as asking us to imagine Alice taller than she *actually* is. One might therefore propose enriching counterpart theory with an 'actually' operator. However, it has long been observed that counterpart theory struggles to analyse the 'actually' operator in a satisfactory manner (Hazen 1979; Fara and Williamson 2005).¹⁸ According to many, accounting for (52) within counterpart cannot be done—or at least faces substantial challenges. One might hope for an analysis of sentences like (52) that avoids the thorny issues raised by analysing 'actually' within counterpart theory.

Russell (1905) and Milne (1993) suggest enriching the language with degrees. Indeed, this is the standard approach in work on the semantics of comparatives (Kennedy 2005). Lewis (1986, 13) is unsatisfied with this reply for sentences that do not overtly refer to degrees, wishing to stick closer to the sentence's surface form. But introducing degrees seems to get the counterpart reply out of a great deal of trouble, by avoiding the need for an 'actually' operator. Since counterpart relations and *de re/de dicto* readings are separate components, introducing degrees would allow the counterpart reply to avail of *de re* readings of degrees (such as the size of the yacht, or Alice's height) when needed. Such readings are required anyway to account for examples such as (29) and (31b).

By incorporating *de re* readings in this way, the counterpart reply can overcome the challenges facing the *de re* reply. A central problem with the *de re* reply was that counterfactual antecedents appear to behave differently even on their *de dicto* readings. By introducing counterpart relations over and above *de re* readings, the counterpart reply can propose that both terms are read *de dicto*, while the differing behaviour is due to variation in counterpart strictness (loose for subjects, strict for objects). This allows the counterpart reply to account for (37) (overcoming Challenge 1), and is compatible with object terms in counterfactual antecedents being interpreted *de dicto* (overcoming Challenges 2, 3, and 5). Furthermore, one may assume that the preference for the choice of a counterpart relation is not subject to the principle of charity (overcoming Challenge 4).

¹⁷Such examples (52) were first discussed in the context of counterpart theory by Milne (1993). For responses, see Forbes (1994), Wehmeier (2012), Kocurek (2016), and Borisov (2023).

¹⁸For attempts to solve the issues, see Meyer (2013), Russell (2013), Rigoni and Thomason (2014), and Bacon (2014). For further problems, see Milford (2024). Lewis, for his part, seems not to be bothered by the fact that quantified modal logic struggles to represent counterpart-theoretic notions (Lewis 1971, 206, n. 4, 1986, 13–17).

4.2 Challenges for the counterpart reply

Even when extended in this way, there are a number of reasons to be unsatisfied with the counterpart reply.

To start, there are the usual, overarching criticisms of counterpart theory. Critics have argued that it gives rise to a deviant modal logic (Hazen 1979; Kripke 1972; Forbes 1982; Woollaston 1994; Cresswell 2004; Ramachandran 2008). For example, it validates the controversial Converse Barcan Formula $\Box \forall x Fx \rightarrow \forall x \Box Fx$ but invalidates simple, intuitive rules such as $\Box (Fa \wedge Gb) \rightarrow \Box Fa$, Leibniz' Law, and the necessity of identity (Kripke 1972; Plantinga 1974; Hall, Rabern, and Schwarz 2024). These consequences may seem so unpalatable that one might prefer to avoid counterpart theory in general.

Second, we would have to give up much of what we thought we knew about counterpart theory. We would like to have some independent grasp on the counterpart relation. Lewis offers an answer: the counterpart relation is “based on similarity” (1973, 41). For Lewis, “something has for counterparts at a given world those things existing there that resemble it closely enough in important respects of intrinsic quality and extrinsic relations, and that resemble it no less closely than do other things existing there” (1973, 39).

These are eminently plausible ideas. However, nothing in them leads us to expect that subject and object terms should differ in what counterpart relations they summon up. For instance, given that Alice and Bob are in different locations, worlds where they are where Alice currently is would appear to be equally similar to the actual world as worlds where they are both where Bob currently is. If the counterpart relation is based on similarity, we would therefore expect the counterpart relation to not distinguish between the antecedents “if Alice were near Bob” and “if Bob were near Alice”. Moreover, it seems perfectly clear who Alice and Bob's counterparts are in each of these worlds: the people with all the same properties as Alice and Bob in the actual world, except for a possible difference in location. Lewis's proposal would lead us to believe that when we interpret an antecedent such as “if Alice were near Bob”, we temporarily suspend belief that people who are exactly like Bob, with the same history and intrinsic properties, except for being in a different location, count as Bob's counterparts.

Lewis does not offer any indication why the counterpart relations would differ between subject and object terms in this way, admitting that “I cannot say in general how grammar and context control which counterpart relation is used where” (Lewis 1973c, 436). It is reasonable to find this unsatisfying, and to therefore seek an alternative account on which differences between subject–object are expected.

5 The aboutness reply

The fourth reply we consider is to add a notion of *aboutness*, or *subject matter* to the theory of conditionals. We will implement this using McHugh's aboutness theory of counterfactuals from

McHugh (2022, 2023).¹⁹

The key tenet of the aboutness theory is that when we interpret a counterfactual antecedent, we allow what the sentence is *about* to vary, and fix what it is not about. The theory interprets counterfactuals using the following steps (for further details and a formalisation see McHugh 2022, 2023, 63ff.).

1. Pick a time at which to imagine the change, called the *intervention time*.²⁰
2. Allow the part of the world the antecedent is about at intervention time to vary.
3. Play forward the laws.
4. Stick on the actual past.
5. Restrict to worlds where the antecedent holds.
6. A counterfactual is true just in case its consequent is true at all of/the selected resulting world(s).²¹

On this approach we split the world into *foreground* and *background*, allowing the foreground to vary while fixing the background. The background is, in other words, the *ceteris in ceteris paribus*—the “all else” in “all else being equal”.²²

There is a striking parallel between the foreground–background distinction, as it appears in the semantics of counterfactuals, and the figure–ground distinction, as studied by Gestalt

¹⁹An earlier approach is Ciardelli, Zhang, and Champollion (2018), who make use of the same foreground–background distinction in formulating their background semantics of counterfactuals. They propose that “Background facts are held fixed while making a counterfactual assumption, while foreground facts are allowed to change” (2018, 599). While Ciardelli, Zhang, and Champollion (2018) formulate their account in terms of a foreground–background distinction, their ultimate proposal does not account for the contrasts we have observed, since they predict that logically equivalent sentences have the same foreground and background.

²⁰Other semantics of conditionals that make use of such a time include Thomason (1970), Thomason and Gupta (1980), Kaufmann (2005), Ippolito (2013), Khoo (2015), and Canavotto (2020). Condoravdi (2002) calls this the *temporal perspective*, Arregui (2005) and Ippolito (2013) call it the *accessibility time*, Khoo (2015) calls it the *modal time* and Khoo (2017) calls it the *counterfactual time*.

²¹I write “all of/the selected resulting worlds(s)” to leave open whether counterfactuals quantify universally over the resulting domain or select a unique world from it. For discussion see McHugh (2023, 59–61).

²²An alternative way to add aboutness to a theory of counterfactuals is via an accessibility relation. A number of theories of counterfactuals make use of a contextually-supplied accessibility relation, whereby the antecedent is only evaluated at accessible worlds (see von Fintel 2001; Gillies 2007; Williams 2008; Moss 2012; Mandelkern 2018). Further components of the theory, such as a similarity order or conditional selection function, then determine which accessible worlds make up the counterfactual domain. One may propose that the accessible worlds tend to be those that agree with the actual world on everything the antecedent is not about. The key difference between this proposal and the aboutness theory is that, according to the aboutness theory, aboutness is part of the literal truth conditions of counterfactuals, while on the alternative view just sketched, aboutness merely guides the selection of a context-dependent parameter—the accessibility relation. This makes a difference to which inferences of counterfactual logic end up valid, given the standard view that validity is determined while holding the context fixed (an inference is valid just in case for every context where the premises are true, the conclusion is true). For reasons of space, we will not compare a semantic or pragmatic implementation of aboutness here.

psychologists, Russell, and Talmy.²³ In particular, the idea that we allow the foreground to vary and fix the background parallels how Talmy understands the figure–ground distinction:

The FIGURE object is a moving or *conceptually* movable point whose path or site is conceived as a variable the particular value of which is the salient issue. [...] The GROUND object is a reference-point, having a stationary setting within a reference-frame, with respect to which the FIGURE’s path or site receives characterization.

(Talmy 1975, 419)

We tend to vary the figure and fix the ground.

It is tempting to think that these two distinctions—the figure–ground distinction on the one hand, studied by Gestalt psychologists and Talmy, and the foreground–background distinction on the other, as it applied in the aboutness theory of counterfactuals—are in fact one and the same. Thus if we follow Talmy in saying that in “Alice is near Bob,” by default Alice is the figure and Bob the ground, this gives us an immediate prediction for how counterfactuals behave. We predict that when we put this sentence in a counterfactual antecedent, “if Alice were near Bob,” we vary Alice but not Bob. As we saw in (4), this is precisely what we observe.

There are many choices as to what kinds of thing sentences are about (see Hawke 2018 for an overview). For instance, some take subject matters to be *questions* (Lewis 1988a,b; Plebani and Spolaore 2021), others take them to be *states* (Fine 2017b, 2020), and yet others take them to be individuals (Perry 1989). We need not decide between these options here.

A central question facing theories of aboutness is whether logically equivalent sentences can differ in what they are about. According to intensional analyses of aboutness (such as Putnam 1958 and Goodman 1972), they cannot, while according to hyperintensional theories (Fine 2017b; Hawke, Hornischer, and Berto 2024), they can. Reinhart, in her influential treatment of sentence topics, writes that, “It is a crucial fact about sentence topics, that equivalent sentences may have different topics” (Reinhart 1981, 58).

There is evidence that logically equivalent sentences can differ in what they are about. Consider the following scenario, modelled on one by Fine (2014; see also Goodsell 2022). A simple universe contains one gold coin and infinitely many silver coins. Each coin can face either heads or tails. Compare the sentences:

- (53) a. Infinitely many of the coins are facing tails.
 b. Infinitely many of the silver coins are facing tails.

These sentences are logically equivalent.²⁴ If infinitely many of the silver coins are facing tails, then infinitely many of the coins are, and given that there is only one gold coin, if infinitely many

²³As we saw in section 1, Russell did not conceive of the distinction in terms of figure and ground—terminology that only appeared later—but in terms of subject and assertion (Russell 1903). The differing terminology seems to nonetheless track the same distinction.

²⁴One might reply that (53a) and (53b) are not logically equivalent, since in a world where the gold coin does not exist, and infinitely many of the silver coins are facing heads, (53b) is true but (53a) is either false or undefined. To

of the coins are facing tails, then infinitely many of the silver coins are. There is, however, a clear sense in which the sentences are *about* different things. (53a) is about all of the coins, including the gold coin, while (53b) is only about the silver coins.

The sentences behave differently as counterfactual antecedents. Suppose the coins are spread out throughout the universe and have no influence on one another. Currently, every coin is facing heads. Compare:

- (54) a. If infinitely many of the coins were facing tails, the gold coin would still be facing heads.
 b. If infinitely many of the silver coins are facing tails, the gold coin would still be facing heads.

(53b) is intuitively true, but (54a) is not. Substitution fails. The aboutness approach captures these judgements. We fix the state of the gold coin, keeping it facing heads. In contrast, (54a)'s antecedent is about all of the coins, so when we interpret it we allow the state of all of the coins to vary, including the gold coin.

5.1 The Subject Constraint

The aboutness theory developed here seems ideally suited to account for subject–object contrasts in counterfactuals. There is a long tradition of thinking that the *grammatical* subject of a sentence is also its subject *matter*, that the two notions of subject coincide (Geach 1962; Strawson 1964, 1974).²⁵

Russell also thinks of sentences as being about their grammatical subjects. Russell's his of statements into 'subject' and 'assertion'. In 'A is younger than B', for example, he claims that 'A' is the subject and 'is greater than B' is the assertion. After this claim, he then proposed that equivalent sentences, by structuring information in different ways, can express distinct propositions, and that the difference is one of aboutness.

The proposition "humanity belongs to Socrates," which is equivalent to "Socrates is human," is an assertion about humanity; but it is a distinct proposition. In "Socrates is human," the notion expressed by *human* occurs in a different way from that in which it occurs when it is called *humanity*, the difference being that in the latter case, but not in the former, the proposition is *about* this notion.

(Russell 1903, §48)

For example, in 'Socrates resembles Adonis', the grammatical subject is 'Socrates'. According to this tradition, then, the sentence is about Socrates and not about Adonis.

get around this, we may add to the sentences the information that there is one gold coin and infinitely many silver coins. The resulting sentences will then be logically equivalent.

²⁵See Paul 2010 for an overview. For a discussion of the notion of grammatical subject in English, see Conner (1968, 43ff.) For a list of criteria to identify subjects, see Biber et al. (2021, 127).

Of course, the sentence is not about any property of Socrates whatsoever, but specifically his appearance. This suggests that the predicate should also play a role in determining what the sentence is about: the sentence ‘Socrates resembles Adonis’ is about the state of Socrates’ appearance (on state-based analyses of aboutness), or what Socrates’ appearance is (on question based analyses). This suggests the following, loose characterisation.²⁶ I qualify this as a default, rather than a universal generalisation, for reasons discussed below.

The Subject Default. By default, sentences are only about their subjects.

A sentence of the form *Fa* is, by default, only about how *a* stands with respect to satisfying *F*.

Paired with the aboutness theory, the Subject Default has the resources to account for the subject–object contrasts we have observed. To illustrate, recall (7).

- (7) a. If Socrates resembled Adonis, they would both be handsome.
- b. If Adonis resembled Socrates, they would both be handsome.
- c. If Adonis and Socrates resembled each other, they would both be handsome.

Given that ‘Socrates resembled Adonis’ is about Socrates’ appearance and not about Adonis’, according to the aboutness theory, when we interpret ‘if Socrates resembles Adonis’ we vary Socrates’s appearance and fix Adonis’s properties. We then restrict to sentences where the sentence ‘Socrates resembles Adonis is true’, resulting in worlds where Socrates and Adonis are both handsome. In contrast, given that ‘Adonis resembles Socrates’ is about Adonis’s appearance and not Socrates’, the aboutness theory predicts that we consider worlds where both are not handsome. And given that ‘Adonis and Socrates resemble each other’ is about both of their appearances, the aboutness theory predicts that we allow both of their appearances to vary. When we then restrict to worlds where the antecedent is true, resulting in some worlds where they are both handsome and some worlds where they are both not handsome—the right result.

²⁶Note that the grammatical subject of a sentence is distinct from what is often called a sentence’s *topic* (Reinhart 1981) or, in the Prague School of functional linguistics, its *theme* (Firbas 1964; Mathesius 1975). We can isolate the topic using terms such as *speaking of*, *as for*, and *talking about* (Reinhart 1981), as in (i) and (ii).

- (i) a. Talking about Socrates, if he resembled Adonis, they would both be handsome.
- b. Talking about Adonis, if Socrates resembled him, they would both be handsome.
- (ii) a. Talking about Socrates, if Adonis resembled him, they would both be handsome.
- b. Talking about Adonis, if he resembled Socrates, they would both be handsome.

Forcing the topic in this way does not alter the sentences’ truth value. Intuitively, both sentences in (i) are true, while both sentences in (ii) are false. While the topic and grammatical subject are distinct notions, it is nonetheless true that the grammatical subject will often happen to be the topic. According to Reinhart, “There is a strong preference in discourse to interpret the grammatical subject of the sentence as its topic, or to place the topic in subject position. [...] this preference is only a matter of tendency and we can use sentences with non-subjects as topics” (Reinhart 1981, 62).

For a second example, recall from (12) that when we interpret “if every raven were black” we tend to change the actual ravens’ colour, while keeping them as ravens. One way to account for this is to follow the *de re* reply, interpreting this antecedent as “if every actual raven were black”. Where *a* and *b* are the actual ravens, say, this is equivalent to “if *a* and *b* were black”.

This aboutness approach offers another route to the same result. We may interpret the entire sentence *de dicto*—expressing the proposition true at a world *w* just in case every raven in *w* is black in *w*—and rely instead on the aboutness relation.²⁷ A common thought is that “all ravens are black” is about the actual ravens. This idea is endorsed by Yablo (2014, 2, 19–22). More generally, Matthewson (2001) argues that quantified expressions such as “all ravens” denote the plurality of objects that actually have this property. And Ben-Yami (2012) writes that, where *q* is a quantifier, “In a sentence of the form ‘*q S are P*’ the subject term *S* is used as a plural referring expression, determining the plurality over which we quantify.” Then given the Subject Default, the predicate *is black* tells us that the sentence is about the actual ravens’ colour. According to the aboutness theory, then, when we interpret “if all ravens were black” we vary the actual ravens’ colour.

5.2 The default status of the subject constraint

According to the Subject Default, sentences *tend* to be about their subjects. It is a default, rather than a hard constraint. In this section we consider two cases where this default does not hold.

A first case comes from dummy subjects (such as *it* and *there*).

- (55) a. If it were necessary to call in advance, I would do so.
b. If there were a coin in my pocket, I would spend it.

It is often assumed that dummy subjects are semantically empty: they do not denote anything in particular (Haider 2019). Strictly enforcing the rule that sentences are only about their subjects would imply that (55) are not about anything, and therefore on the aboutness theory that we do not change anything when we interpret them—an undesirable result.²⁸

A second case occurs when imagining a counterfactual antecedent true requires changing more than just the subject. We have already seen cases of this in (23)–(26). When there is a conflict between the truth of the antecedent and changing only the subject’s properties, the truth of the antecedent wins out.

²⁷Note that this reply would not do for antecedents like “if every raven were a parrot” or “if everyone inside the room were outside the room”, which are contradictory *de dicto* but consistent *de re*. In these cases, the aboutness approach needs the resources of the *de re* reply.

²⁸An alternative stance is that dummy subjects refer to a general, loosely specified state; say, the state of it not being necessary to call in advance, or the state of there being no coin in my pocket. Given this response, dummy subjects would no longer challenge the idea that sentences are only about their subjects.

5.3 Challenges for the aboutness theory

This section points out two challenges facing the aboutness theory and sketches some replies.

Disjunctive antecedents. Suppose there are two blocks, A and B, each sitting on a patch of grass, with a path between them (Figure 12). Consider (56).



Figure 12

- (56) a. If A were beside B or B were beside A, both blocks would still be on grass.
 b. If A and B were beside each other, both blocks would still be on grass.

Here we find an interesting contrast. In (56a), we imagine A where B is, or B where A is—we fix the location of one block in each case—and the conditional is true, while in (56b), we allow the locations of both blocks to vary simultaneously and the conditional is not true.

At first glance, the aboutness theory struggles with the contrast, since both blocks appear as subjects in both antecedents. However, there is a reply. A popular view is that disjunctive antecedents are not interpreted as a single antecedent but as multiple antecedents, called the conditional's *alternatives*, and that a conditional is true just in case it holds for each of its alternatives (Alonso-Ovalle 2006, 2009; Ciardelli 2016; Santorio 2018; Willer 2018; Khoo 2022). On this view, the alternatives to a disjunctive conditional antecedent *A or B* are the individual disjuncts, *A* and *B*. This view predicts (56a) to be true just in case the following are true:

- (57) a. If A were beside B, both blocks would still be on grass.
 b. If B were beside A, both blocks would still be on grass.

The aboutness approach readily predicts the truth of these conditionals. Incorporating alternatives, then, (56a) is no longer a challenge to the aboutness theory.

In contrast, an antecedent like 'if A and B were beside each other', which lacks disjunction, has a single alternative: the entire antecedent. Both blocks appear as subjects, so the aboutness approach predicts that when we interpret (56b) we vary the locations of both blocks simultaneously, as desired.

Quantified antecedents. A second potential challenge for the aboutness theory comes from quantified antecedents such as (58).

- (58) *Alice is one of the people on a football field. She is at a goalpost.*
 If everyone on the pitch were less than one metre away from Alice, everyone would be near a goalpost.

When we interpret (58), we move everyone else to Alice’s actual location: the conditional is true. Plausibly, ‘everyone on the pitch is less than one metre from Alice’ is about the locations of everyone on the pitch, including Alice’s location. On the aboutness theory, we would therefore expect to vary everyone’s location, including Alice’s. In fact, we hold her location fixed.

Here again the aboutness theory has a reply: domain restriction. It is well-known that quantifiers come with a covert domain restriction (Westerståhl 1985; von Stechow 1994; Stanley and Szabo 2000), illustrated in (59).

- (59) a. Everyone shook hands with everyone.
 b. Everyone one on the pitch is more than one metre away from Alice.

(59a) typically does not imply that everyone also shook hands with themselves. Likewise, even when Alice is one of the people on the pitch, (59b) typically does not imply that Alice is more than 1m away from herself. The quantifier is restricted to exclude Alice. If the same goes for (58), then its antecedent is not about Alice at all, and the aboutness theory predicts that when we interpret it we hold Alice’s location fixed, as desired.

6 Conclusion

The figure–ground distinction, studied by the Gestalt psychologists, Russell and Talmy, is fundamental to how we structure information around us. A hitherto unnoticed fact about the distinction is that it gives rise to striking contrasts in counterfactuals. The contrasts show up in a wide range of environments, including comparatives (*taller than*), locatives (*near*, *adjacent to*, *on top of*), transitive verbs (*resembles*, *is similar to*), and quantifiers (*every*, *some*, *no*). As Talmy noticed, there are logically equivalent sentences that nonetheless differ in their division of figure and ground: it is a widespread source of hyperintensionality in language, operating along a dimension of meaning beyond truth conditions.

This is a problem for theories of counterfactuals that validate Substitution, such as Stalnaker’s and Lewis’s, which are committed to the idea that logically equivalent sentences are intersubstitutable *salva veritate* in conditional antecedents. Indeed, it is also a challenge to the weaker principle of Exact Substitution and Fine’s semantics of counterfactuals, despite the remarkable fine grain of that approach. For it is plausible that the sentences exhibiting the figure–ground not only have the same truth conditions, but also the same truthmakers. This is supported by Fine’s own arguments, as well as independent arguments from the theory of reciprocals.

The aboutness theory is ideally placed to account for figure–ground contrasts in counterfactuals. It holds that when we interpret a counterfactual antecedent we distinguish foreground from background: the facts we allow to vary from those we hold fixed. This lines up neatly with how Talmy think of the distinction, with the figure “a moving or conceptually moveable point” and the ground having “a stationary setting.” Generalising from Russell and Talmy’s examples, we proposed the Subject Constraint: by default, sentences are only about their subjects. Paired with the aboutness theory, this constraint immediately makes the right predictions for

figure–ground contrasts in conditionals. For example, it predicts that when we interpret “if Socrates resembled Adonis,” we allow Socrates’ appearance to vary and fix Adonis’, while this is reversed with “if Adonis resembled Socrates,” and given “if Socrates and Adonis resembled each other” we allow the appearance of both to vary.

There is a response available to proponents of (Exact) Substitution: the *de re* reply. It alleges that comparators are anchored to their values in the actual world; for instance, that “if Socrates resembled Adonis” is read as “if Socrates’ appearance resembled Adonis’ appearance in the actual world.” While this results in plausible truth conditions, it fails to predict the correct range of attested readings. Specifically, it predicts there to be true *de dicto* readings where in fact we observe none. Ruling out the *de dicto* readings entirely does not account for the behaviour of complex conditional antecedents and belief reports. It moreover violates what we know about the *de re/de dicto* ambiguity in general, and the principle of charity in particular. In contrast, on the aboutness theory these difficulties do not arise, since it accounts for figure–ground contrasts regardless of whether the antecedents in question are read *de dicto* or *de re*.

There are many questions to ask about the aboutness theory. How we decide in general what parts of the world sentences are about? What other factors, beyond the sentence’s subject, determine the aboutness relation? One may appeal to truthmakers here (Fine 2017c), arguing that a sentence *A* is about state *s* just in case *s* exactly verifies or falsifies *A*, with the additional default constraint that *s* be part of the subject of *A* to comply with the Subject Constraint. These are rich open questions, though we hope to have shown that they are questions worth asking: that aboutness is a worthy primitive in the theory of counterfactuals, one superior to the notion of similarity in its account of figure–ground contrasts. For the last half-century, theories that limit themselves to the truth conditions of counterfactual antecedents have been at the foreground of work on counterfactuals. It is time for these theories to move to the background, and for the hyperintensional notion of aboutness to take their place.

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